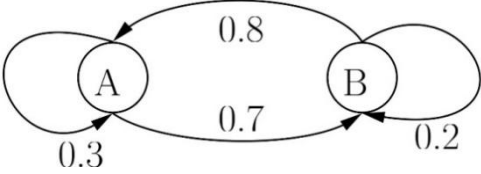




Date: 04/06/2025	CIE - 2	Max. Marks:100
Semester: VI	UG	Duration: 2 Hrs. (2pm-4pm)
Course Title: Mathematical Modelling		Course Code: MA266TEU

Department of Mathematics

	Quiz			
SL. No	Questions	M	BTL	CO
1	The Fibonacci Numbers 1, 1, 2, 3, 5, 8, 13, 21,... can be modelled in a difference equation of the form _____.	2	2	2
2	Solve $u_{t+1} = 8x_t$, $x_0 = \frac{1}{2}$.	2	1	1
3	With respect to a characteristic brown eyes, suppose the proportions of the dominants, hybrids and recessives in 9 th generation are p_9 , q_9 and r_9 respectively, then $r_{10} =$ _____, and $q_{10} =$ _____.	2	1	1
4	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	2	3	2
5	Find the fixed probability vector of the stochastic matrix $A = \begin{bmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{bmatrix}$.	2	3	3
SL. No	Test	M	BTL	CO
1.	Develop a model through difference equations in genetics based on Hardy-Weinberg law.	10	2	2
2.	Discuss Cobweb model about the variation of price and quantity of a commodity produced by a farm.	10	2	2
3.	Prove that the Markov chain with transition probability matrix $\begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is irreducible. Also find its stationary probability vector.	10	2	2



4. a	Suppose a person borrows a sum S_0 at compound interest i and wants to 'amortize' his debt and S_t be the amount due at the end of t years. Model this in the form of a difference equation and find the amortization factor.	6	1	1
4. b	A college professor drives or takes the college bus to reach the college. If he drives to work, then the next day he takes the bus with the probability 0.2. On the other hand if he takes the bus, then the next day he drives with probability 0.3. Find the transition matrix for the chain of mode of transport.	4	2	3
5.	Consider a network containing five individuals viz. A, B, C, D and E. The system A can communicate to B and C; B can communicate to A and C, C can communicate to A, B and D; D can communicate to C and E; and E can communicate to B only. Assuming that if a person can communicate with n people, he will send a message to any one of them with probability $1/n$, rank the five people in accordance with their importance in the network in the long run.	10	3	4

**** ALL THE BEST ****

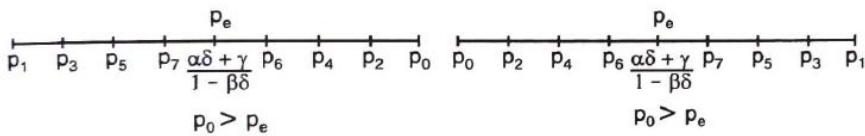
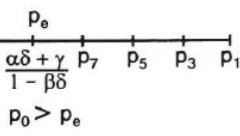


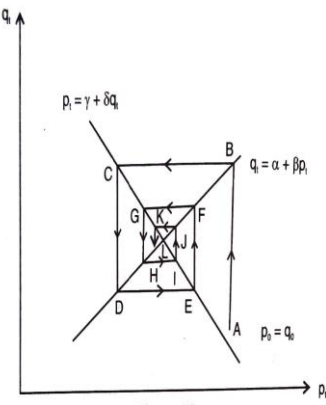
Date: 04/06/2025	CIE - 2	Max. Marks:100
Semester: VI	UG	Duration: 2 Hrs. (2pm-4pm)
Course Title: Mathematical Modelling		Course Code: MA266TEU

SCHEME & SOLUTIONS

S No	Solutions with Scheme	Marks
	Part-A	
1.	$x_{n+1} = x_n + x_{n-1}$	1+1
2.	$x_t = 8^t \cdot \frac{1}{2}$	1+1
3.	r_9, q_9	1+1
4.	$P = \begin{bmatrix} 0.3 & 0.7 \\ 0.8 & 0.2 \end{bmatrix}, \pi_A = \frac{8}{15}, \pi_B = \frac{7}{15}$	1+1
5.	$\pi = \begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix}$	2

Q.No	Part-B	Marks
1	Every gene occurs in two forms, dominant (denoted by 'G') and a recessive ('g'). Thus with respect to a characteristic, an individual may be a dominant (GG), a hybrid (Gg or gG) Or a recessive. In the 'n'th generation, let the proportions be p_n, q_n and r_n so that $p_n + q_n + r_n = 1, p_n, q_n, r_n \geq 0$. We assume that individuals in this generation mate at random. p_{n+1} = The probability that an individual in the $(n + 1)th$ generation is dominant (GG) = (Probability that this individual gets a G from father $\times$ Probability that the individual gets a G from mother) $= \left(p_n + \frac{1}{2}q_n\right)\left(p_n + \frac{1}{2}q_n\right) = \left(p_n + \frac{1}{2}q_n\right)^2$ Similarly, $q_{n+1} = \left(p_n + \frac{1}{2}q_n\right)\left(r_n + \frac{1}{2}q_n\right) + \left(r_n + \frac{1}{2}q_n\right)\left(p_n + \frac{1}{2}q_n\right) = 2\left(p_n + \frac{1}{2}q_n\right)\left(r_n + \frac{1}{2}q_n\right)$ $r_{n+1} = \left(r_n + \frac{1}{2}q_n\right)^2$ So that $p_{n+1} + q_{n+1} + r_{n+1} = \left(p_n + \frac{1}{2}q_n\right)^2 + 2\left(p_n + \frac{1}{2}q_n\right)\left(r_n + \frac{1}{2}q_n\right) + \left(r_n + \frac{1}{2}q_n\right)^2 = 1$ Similarly, $p_{n+2} = \left(p_{n+1} + \frac{1}{2}q_{n+1}\right)^2 = \left[\left(p_n + \frac{1}{2}q_n\right)^2 + \left(p_n + \frac{1}{2}q_n\right)\left(r_n + \frac{1}{2}q_n\right)\right]^2$ $= \left[\left(p_n + \frac{1}{2}q_n\right)\left(p_n + \frac{1}{2}q_n + r_n + \frac{1}{2}q_n\right)\right]^2 = \left[\left(p_n + \frac{1}{2}q_n\right)^2 (p_n + q_n + r_n)^2\right] = \left(p_n + \frac{1}{2}q_n\right)^2 = p_{n+1}$ and	2+2+ 2+2+2

	$q_{n+2} = q_{n+1}, r_{n+2} = r_{n+1}$ This means that the proportions of dominants hybrids and recessives in the $(n + 2)th$ generation are same as in the $(n + 1)th$ generation. Thus in any population in which random mating takes place with respect to a characteristics the proportions of dominants, hybrids and recessive do not change after the first generation. This is known as Hardy-Weinberg law after the mathematician Hardy geneticist Weinberg who jointly discovered it.	
2	Let p_t be the price of a commodity in the year t and let q_t be the amount of the commodity available in the market in the year t. The following assumptions are made: i) Amount of the commodity produced this year and available for sale is a linear function of the price of the commodity in the last year. $q_t = \alpha + \beta p_{t-1}$ ----(1) Where, $\beta > 0$, since if the last year's price was high, the amount available this year will also be high. ii) The price of the commodity this year is a linear function of the amount available this year. $p_t = \gamma + \delta q_t$ ----(2) Where $\delta < 0$, since if q_t is large, the price would be low. From (1) AND (2), $p_t = \gamma + \delta(\alpha + \beta p_{t-1}) = \gamma + \delta\alpha + \beta\delta p_{t-1} = (\beta\delta)p_{t-1} + (\gamma + \alpha\delta)$ Which is a non- homogenous difference equation , Its solution is $p_t = (\beta\delta)^t p_0 + \sum_{i=1}^t (\beta\delta)^{t-i} (\gamma + \alpha\delta) = (\beta\delta)^t p_0 + (\gamma + \alpha\delta) [(\beta\delta)^{t-1} + (\beta\delta)^{t-2} + \dots + (\beta\delta) + 1]$ $= (\beta\delta)^t p_0 + (\gamma + \alpha\delta) \left[\frac{1 - (\beta\delta)^t}{1 - \beta\delta} \right] = (\beta\delta)^t p_0 + \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} (\beta\delta)^t$ $p_t - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} = \left[p_0 - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} \right] (\beta\delta)^t$ -----(3) $p_{t-1} - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} = \left[p_0 - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} \right] (\beta\delta)^{t-1}$ -----(4) From (3), $p_t - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} = \left[p_0 - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} \right] (\beta\delta)^{t-1} (\beta\delta) = \left[p_{t-1} - \frac{(\gamma + \alpha\delta)}{1 - \beta\delta} \right] (\beta\delta)$ Since $(\beta\delta)$ is negative, $p_0, p_1, p_2, \dots$ are alternately greater and less than $\frac{(\gamma + \alpha\delta)}{1 - \beta\delta}$. If $\beta\delta > 1$, then the deviation of p_t from $\frac{(\gamma + \alpha\delta)}{1 - \beta\delta}$ goes increasingly. If $\beta\delta < 1$, then the deviation of p_t from $\frac{(\gamma + \alpha\delta)}{1 - \beta\delta}$ goes on decreasingly and ultimately $\cdot p_t \rightarrow \frac{(\gamma + \alpha\delta)}{1 - \beta\delta}$ as $t \rightarrow \infty$.  (a)  (b) Figure (a) and (b) show how the price approaches equilibrium price $p_e = \frac{(\gamma + \alpha\delta)}{1 - \beta\delta}$ as t increases in two cases when $p_0 > p_e$ and $p_0 < p_e$ respectively. In the same way eliminating p_t from equation (1) and (2), we get $q_t = \alpha + \beta\gamma + \beta\delta q_{t-1}$ which has the solution $q_t - \frac{(\alpha + \beta\gamma)}{1 - \beta\delta} = \left[q_{t-1} - \frac{(\alpha + \beta\gamma)}{1 - \beta\delta} \right] (\beta\delta)$ So that q_t also oscillates about the equilibrium quantity q_e.	1+2 +2+2 +1+2

		
3	In P^2, all the entries are positive. So P is irreducible. Stationary probability vector = $V = [\frac{1}{3}, \frac{10}{27}, \frac{8}{27}]$	4 6
4a	If a sum s_0 is invested at compound interest i per unit amount per unit time and s_t is the amount at the end of time t, then we have the difference equation $s_{t+1} = s_t + is_t = (1 + i)s_t$ Which has the solution $s_t = s_0(1 + i)^t$ Which is well-known formulae for compound interest. Suppose a person borrows a sum S_0 at compound interest i and wants to 'amortize' his debt ie., pay the amount and the interest back by payment of n equal installments say R. Let S_t be the amount due at the end of t years, then we have the difference equation $s_{t+1} = s_t + is_t - R = (1 + i)s_t - R$ which is a first order non-homogeneous difference equation. Its solution is $s_t = (1 + i)^t s_0 - \sum_{k=1}^t (1 + i)^{t-k} R = (1 + i)^t s_0 - R \{ (1 + i)^{t-1} + (1 + i)^{t-2} + \dots + (1 + i) + 1 \}$ $= (1 + i)^t s_0 - R \left[\frac{(1+i)^t - 1}{1+i-1} \right]$ At the end of n years $s_n = 0$ Ie., $0 = (1 + i)^n s_0 - \frac{R}{i} [(1 + i)^n - 1]$, $\frac{R}{i} [(1 + i)^n - 1] = (1 + i)^n s_0$ $R = \frac{i(1+i)^n s_0}{(1+i)^n - 1} = s_0 \frac{i}{1 - (1+i)^{-n}} = s_0 \frac{i}{a_n i}$, $a_n i$ is called the amortization fraction in the present value of annuity of 1 per unit time for n periods at an interest rate i.	2 +2+2
4b	$P = \begin{bmatrix} 0.3 & 0.7 \\ 0.8 & 0.2 \end{bmatrix}$	4
5	TPM = $\begin{bmatrix} 0 & 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 1/3 & 1/3 & 0 & 1/3 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$, $VP = V, V = \left[\frac{14}{57}, \frac{16}{57}, \frac{6}{19}, \frac{2}{19}, \frac{1}{19} \right]$ or $V = \left[\frac{14}{57}, \frac{16}{57}, \frac{18}{57}, \frac{6}{57}, \frac{8}{57} \right]$ C is the most important individual in the network.	5 4+1



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SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Date: 02/05/2025	Test - 1	Max. Marks:100
Semester: VI	UG	Duration: 2 Hrs. (2pm-4pm)
Course Title: Mathematical Modelling		Course Code: MA266TEU

Declaration by the Course handling faculties:

As a teaching faculty of the course: **Mathematical Modelling** with code: **MA266TEU**

We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analysing, evaluating, and creating.	
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	
5	Course handling faculty (as applicable) who are responsible for preparing the question paper, Scheme and Solutions have unanimously agreed to utilize this Question paper for conducting the Continuous Internal Evaluation.	
6	The Question paper, Scheme and complete Solutions have been submitted to the Test coordinators within the designated time-frame to ensure the smooth conduction of Continuous Internal Evaluations	

Course handling Faculties:

Name:

Signature

1. Dr. Y. Sailaja

2. Dr. Nivya Muchikel



To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme & complete Solutions in the format	10	
	Total points	50	
Any other comments by the scrutinizer :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	Signature of Course coordinator
Accepted/Rejected	Signature of Scrutinizer (Name:)

Signature of HOD